OGHENEJABOR LAURENCE JABORO

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PROFILE

Final-year chemistry undergraduate at the University of Benin working on de novo protein design, AI-driven small-molecule design and antimicrobial resistance. Co-founder and President of Protein Design for Africa (PDFA), a pan-African non-profit training young scientists in computational protein design. Designed an experimentally validated de novo binder to the cancer target RBX-1 (K_D = 185 nM by BLI), and three AI-designed TEM-1 β-lactamase inhibitors now synthesised and entering biological testing. Leads PDFA's EGFR, TNF-α, GLP-1R and TfR1 campaigns in the Anthropic × Adaptyv Bio Protein Design Competition.

EDUCATION

B.Sc. Chemistry — University of Benin (UNIBEN), Nigeria 2023 – 2027 (expected)

400 Level (final year). Project: computational design of small-molecule inhibitors of TEM-1 β-lactamase (supervisor: Dr. Osaro Iyekowa).

RESEARCH EXPERIENCE

De Novo EGFR Binder Design — Anthropic × Adaptyv Bio Protein Design Competition 2026

Team lead, Protein Design for Africa (Track 2: Claude Max access + $500 Modal credits)

- Led Challenge 1: conditional de novo binders to EGFR, engineered for human/mouse cross-reactivity and pH selectivity. Generated candidates with BindCraft and BoltzGen, refolded with Boltz-2 on Modal, ranked by ipSAE/ipTM; 20 designs submitted, wet-lab results from Adaptyv Bio due November 2026.
- Also leading Challenge 2 (pH-selective, cross-reactive TNF-α miniproteins via BindCraft2), Challenge 3 (GLP-1R agonist miniproteins) and the TfR1 conditional-binder challenge.

De Novo RBX-1 Binder Design — Adaptyv Bio RBX-1 Competition 2026

Experimentally validated binder, K_D = 185 nM by BLI

- Designed 11 de novo binders to RBX-1, the RING-box subunit of Cullin-RING E3 ligases and a target with no known small-molecule binders, using BoltzGen and BindCraft against the N-terminal IDR and C-terminal RING domain.
- Three designs expressed in a validation run sponsored by STEAMulater LLC; the C-terminal RING-domain design bound with K_D = 185 nM (k_{on} 5.5×10^4 M^{-1} s^{-1}, k_o^{ff} 9.0×10^{-3} s^{-1}; ipSAE 0.93, ipTM 0.918).
- Abstract accepted for ABRCMS 2026 (Seattle); result earned PDFA a $5,000 Boltz API credit award.

Computational Design of TEM-1 β-Lactamase Inhibitors — Final-Year Project, UNIBEN 2026 – present

- Designed 1,000 small-molecule inhibitors of TEM-1 β-lactamase (PDB 1BTL) with BoltzMol-1 constrained to Enamine REAL chemical space; independently re-scored all 1,000 with Boltz-2.1 and ranked by binding confidence, ADMET profile and structural alerts.
- Molecular docking in AutoDock Vina and 48-endpoint ADMET profiling with ADMET-AI on Vecura, cross-checked against ADMETLab 3.0; molecular dynamics simulations on Vecura in progress.
- Three lead compounds synthesised by OnePot.ai with LC-MS and ^{1}H NMR characterisation; TEM-1 enzyme inhibition (nitrocefin), disk-diffusion and checkerboard MIC assays scheduled before the end of 2026.

Lassa Fever Antibody Design — Protein Design for Africa 2026

- ~30,000 antibodies designed against the Lassa virus glycoprotein with BoltzProt-1; funded by the Edo State Ministry of Health, with 100 designs planned for wet-lab testing.

Analytical Chemistry Student Researcher — UNIBEN 2025

- Liquid–liquid and solid–liquid extraction, column-chromatography clean-up and GC analysis of TPH and PAH in water and soil samples.

Intern — Centre for Forensic Programmes and DNA Studies (CEFPADS), UNIBEN 2025

- Six-month forensic programme: fingerprint lifting, DNA extraction, PCR and gel electrophoresis.

LEADERSHIP

Co-founder & President — Protein Design for Africa (PDFA) Feb 2026 – present

- Lead a non-profit teaching de novo protein design, with members in six African countries; direct campaigns in binder, antibody and small-molecule design.
- Organised PDFA's inaugural webinar with The Protein Society (June 2026); presented to the Edo State Ministry of Health and secured funding for the Lassa fever programme.

Founding President — ACS International Student Chapter, University of Benin 2026 – present

- Founded the chapter (chartered July 2026) and built a campus community around chemistry research, training and outreach.

Director of Research & Development — Student Chemical Society of Nigeria, Edo State Chapter 2025 – present

- Coordinate academic projects, workshops and training; liaise between student research teams and faculty advisors.

PRESENTATIONS

- ABRCMS 2026, Seattle (accepted): De Novo Protein Design Targeting RBX-1: From Computation to Experimental Validation (poster).

PROFESSIONAL MEMBERSHIPS

American Chemical Society • The Protein Society • British Pharmacological Society • American Society for Microbiology (Global Outreach Student Member) • British Society for Antimicrobial Chemotherapy

TECHNICAL SKILLS

Protein design: BindCraft/BindCraft2, BoltzGen, BoltzProt-1, RFdiffusion, ProteinMPNN; structure prediction with AlphaFold, ESMFold and Boltz-2; ipSAE/ipTM-based ranking; antibody design.

Small-molecule design: BoltzMol-1, Boltz-2.1 re-scoring; molecular docking (AutoDock Vina, PyRx); ADMET prediction (ADMET-AI on Vecura, ADMETLab 3.0, SwissADME); molecular dynamics (GROMACS, OpenMM); RDKit, PyMOL, BIOVIA Discovery Studio Visualizer.

Computing: Python, RDKit, Windows PowerShell, Modal, Boltz API, Vecura, large-scale design pipelines.

Laboratory: Gas chromatography, liquid–liquid and solid–liquid extraction, column chromatography, DNA extraction, PCR, gel electrophoresis.

VOLUNTEERING

Assistant Team Lead, Protocols & Logistics, EBL 8.0 (2026) • Logistics Team, TEDx UNIBEN (2025) • Junior Chamber International Nigeria, UNIBEN • Directorate of Sustainable Development Goals

LANGUAGES

English — C1 (Advanced)

REFEREES

Dr. Anthony E. Aiwonegbe, Associate Professor, University of Benin — anthony.aiwonegbe@uniben.edu, +234 805 236 7370

Dr. Osaro Iyekowa, Final-year project supervisor, Department of Chemistry, University of Benin — osaro.iyekowa@uniben.edu, +234 805 673 3021